

Project Initialization and Planning Phase

Project Title: PocketSmart AI: Your Smart Budget & Recommendation Assistant

Maximum Marks: 5 Marks

Project Proposal (Proposed Solution) Report

The proposal aims to simplify budget-based planning and product selection using a GenAI-powered recommendation system. PocketSmart AI analyzes user preferences, budgets, and contextual needs to generate personalized suggestions across home interiors, party planning, and jewelry selection. The system connects recommendation workflows with multiple product and service platforms.

Project Overview

Objective — To develop an AI-based assistant that understands user budgets and preferences and provides suitable, personalized product or service recommendations.

Scope — The project covers Home Interior Budget Planning, Party Budget Planning, and Jewelry Budget Planning, including optional image-based analysis for jewelry recommendations.

Problem Statement

Description — Managing budgets for home decoration, events, and jewelry shopping can be difficult because users must compare many products, platforms, prices, and options.

Impact — The difficulty of comparing choices can make planning time-consuming and may make it harder for users to stay within their intended budget.

Proposed Solution

Approach — PocketSmart AI uses Generative AI with a FastAPI-based backend to process budget, preferences, contextual requirements, and optional images. Gemini 1.5 Flash Pro is used to analyze the input and generate structured, budget-aware recommendations. The system supports planners for Home, Party, and Jewelry categories.

Key Features

•	AI-based budget analysis and personalized recommendations.
•	Home Interior Planner for furniture, decor, lighting and related items.
•	Party Planner for catering, venue, decoration and entertainment suggestions.
•	Jewelry Planner with occasion, style and optional outfit-image inputs.
•	Recommendations connected to platforms such as Amazon, Flipkart, IKEA, Swiggy, Zomato and OYO.
•	User registration, login, session management and recommendation history.
•	Responsive frontend with structured recommendation displays.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Laptop / PC
Memory	RAM	8 GB or above
Storage	Project files, application and data	256 GB SSD or above
Software	Programming Language	Python
Framework	Backend Framework	FastAPI
AI	Generative AI Model	Gemini 1.5 Flash Pro
Frontend	User Interface	HTML, CSS, JavaScript, Jinja2

Resource Type	Description	Specification / Allocation
Development Environment	IDE	VS Code
Libraries	Application Libraries	FastAPI, Uvicorn, python-dotenv and related libraries